

Student Name:
Student ID:



Nonlinear Finite Element Method (Spring 2025)

Final Assignment

Due Date: 12:00pm, June 6, 2025

1. **Problem 1** (50%) Implement the explicit Finite Element method in Matlab for stress analysis of 2D dynamic finite deformation problems by completing the predefined Matlab functions. Suppose the material follows the J2 isotropic plasticity law with isotropic hardening, the yield stress is given by,

$$\sigma_c = \sigma_0(1 + A\bar{\epsilon}_p)^\alpha$$

where σ_c , σ_0 , A , $\bar{\epsilon}_p$ and α are the yield stress, the initial yield stress, a constant, effective plastic strain, and power, respectively.

Using the Hencky strain in the constitutive law instead of the deformation gradient, i.e.,

$$\epsilon = \frac{1}{2} \log C$$

where C is the right Cauchy-Green deformation tensor. The total strain of the material may be decomposed additively into the elastic and plastic part,

$$\epsilon_{ij} = \epsilon_{ij}^e + \epsilon_{ij}^p$$

and the incremental effective plastic strain is

$$\Delta\bar{\epsilon}_p = \sqrt{\frac{2}{3} \Delta\epsilon_{ij}^p \Delta\epsilon_{ij}^p}$$

Finally, the stress may be calculated by

$$\sigma_{ij} = K\epsilon_{kk}\delta_{ij} + 2G\epsilon_{ij}^{e, \text{dev}}$$

where K and G are the bulk modulus and shear modulus of the material, respectively. The deviatoric part of the elastic strain tensor is given by

$$\epsilon_{ij}^{e, \text{dev}} = \epsilon_{ij}^e - \frac{1}{3}\epsilon_{kk}\delta_{ij}$$

- (a) Functions in *main_dynamics.m*, *g_center.m*, *plot_dyn_results.m*, *quadplot.m*, and *shape_function.m* are completed. *main_dynamics.m* is the main program of the Matlab code, which shows the steps for a Finite Element analysis process. Read the code starting from *main_dynamics.m* to understand the meaning of data structures and follow the calling order of the subroutines.

- (b) Implement the functions marked with "TODO" comments following the instructions in the code:

`generate_mesh.m`, generate the mesh, i.e., node coordinates and element connectivity table
`lumped_mass.m` : formation of the lumped mass matrix
`consistent_mass.m` : formation of the consistent mass matrix
`boundary_coditions.m` : set up the displacement boundary conditions and initial conditions
`time_step.m` : determine the critical time step of the analysis
`explicit_solver.m` : central difference time integrator
`constitutive.m` : compute the stress according to J2 isotropic plasticity law
`fint_vector.m` : formation of the internal nodal force vector

Note that the important data structures defined in the Matlab code are

$$x_a = \begin{pmatrix} x^{(1)} & y^{(1)} \\ x^{(2)} & y^{(2)} \\ \vdots & \vdots \\ x^{(N)} & y^{(N)} \end{pmatrix}, \quad elem = \begin{pmatrix} I^{(1)} & J^{(1)} & \dots & K^{(1)} \\ I^{(2)} & J^{(2)} & \dots & K^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ I^{(M)} & J^{(M)} & \dots & K^{(M)} \end{pmatrix}, \quad xg = \begin{pmatrix} x^{(1)} & y^{(1)} \\ x^{(2)} & y^{(2)} \\ \vdots & \vdots \\ x^{(M)} & y^{(M)} \end{pmatrix}.$$

where $x^{(i)}$ and $y^{(i)}$ in x_a are the coordinates of node $i \in \{1, 2, \dots, N\}$, $\{I^{(q)}, J^{(q)}, \dots, K^{(q)}\}$ are the indices of the nodes of element $q \in \{1, 2, \dots, M\}$, i.e. $elem$ is the connectivity table, and $x^{(q)}$ and $y^{(q)}$ in xg are the coordinates of the barycenter of element q .

2. **Problem 2** (50%) Consider a dynamic elastoplastic problem on the rectangle panel domain as shown in Figure 1. There is no vertical displacement on the bottom and no horizontal displacement

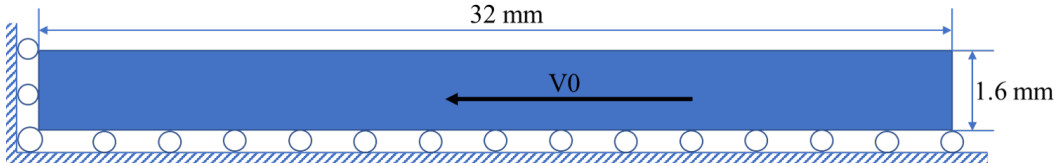


Figure 1: Taylor bar configuration.

on the left. All other edges are traction free, i.e. $\bar{\mathbf{t}} = \mathbf{0}$. Plane strain condition is considered. An initial velocity v_0 in the left direction is applied to the entire domain at time $t = 0$. Material properties for the panel are listed in Table 1.

Young's modulus (GPa)	Poisson ratio	Density (Kg/m ³)	Initial yield stress σ_0 (MPa)	Constant A	Power α
100	0.3	8950	400	10	0.25

Table 1: Material properties

- (a) Use your matlab code developed in Problem 1 to solve the problem with initial conditions $V_0 = 200$ m/s. Plot the deformed panel showing the effective plastic strain, the pressure, and the von-mises stress distribution of the domain at time $t = 5, 10, 15$ and $20 \mu\text{s}$ in your report. Plot the stress-strain curve at an element which experienced plastic deformation.

- (b) Solve this problem in Abaqus or other commercial software and validate your solution from (a) against the one obtained in commercial software by comparing the von-mises stress distribution, effective plastic strain, pressure and displacement field.